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Exam questions

Principal Component Analysis (PCA): Method, Calculation and Interpretation

Question

Explain in detail the principal component analysis (PCA) method, including the assumptions, calculation and interpretation of this technique.

Introduction and Objectives

Principal Component Analysis (PCA) is a dimensionality reduction method that transforms a set of potentially correlated variables into a set of linearly uncorrelated variables called **principal components**.

Main Objectives

- Reduce data dimensionality while preserving maximum information
- Identify directions of greatest variance in the data
- Decorate variables
- Visualize multidimensional data

Fundamental Assumptions

Statistical Assumptions

1. **Linearity**: Relationships between variables are linear
2. **Normal Distribution**: Data approximately follows a multivariate Gaussian distribution

$$X_i \sim \mathcal{N}(\overline{\mathbf{x}}, \Sigma)$$

3. **Importance of Variance**: Variance is an appropriate measure of the information contained in the data
4. **Orthogonality**: Principal components are orthogonal to each other

Data Prerequisites

- Data must be **centered** (zero mean)
- Variable scales must be comparable (normalization recommended)
- No significant missing values

Calculation of Principal Components

Step 1: Data Preparation

Centering: Subtract the mean of each variable

$$\mathbf{x}_i^{\text{centered}} = \mathbf{x}_i - \overline{\mathbf{x}}, \quad \text{where} \quad \overline{\mathbf{x}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{x}_i$$

Normalization (optional but recommended): Divide by the standard deviation

$$\mathbf{x}_i^{\text{norm}} = \frac{\mathbf{x}_i^{\text{centered}}}{\sigma}, \quad \text{where} \quad \sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \overline{x})^2}$$

Step 2: Calculation of the Covariance Matrix

For N observations with D variables, the covariance matrix Σ is:

$$\Sigma = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{x}_i - \overline{\mathbf{x}})(\mathbf{x}_i - \overline{\mathbf{x}})^\top$$

Properties of Σ:

- Symmetric: $\Sigma = \Sigma^\top$
- Positive semi-definite: $\mathbf{v}^\top \Sigma \mathbf{v} \geq 0$ for all $\mathbf{v}$
- Diagonal elements: variances of variables
- Off-diagonal elements: covariances between variables

Step 3: Eigenvalue Decomposition

We seek the eigenpairs $(\lambda_d, \mathbf{u}_d)$ such that:

$$\Sigma \mathbf{u}_d = \lambda_d \mathbf{u}_d, \quad d = 1, \dots, D$$

Order of eigenvalues: $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_D \geq 0$

Matrix of eigenvectors: $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_D]$

Fundamental Relation: Eigenvalue = Variance

Mathematical Demonstration

Step 1: Projected variance

Let $\mathbf{u}$ be a unit vector. The projection of an observation $\mathbf{x}_i$ onto $\mathbf{u}$ is:

$$p_i = \mathbf{u}^\top \mathbf{x}_i$$

Step 2: Variance calculation

The mean of projections (for centered data) is zero:

$$\bar{p} = \mathbf{u}^\top \bar{\mathbf{x}} = 0$$

The variance of projections is therefore:

$$\mathrm{Var}(\mathbf{u}) = \frac{1}{N}\sum_{i=1}^{N}(p_i)^2 = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{u}^\top \mathbf{x}_i)^2$$

Step 3: Matrix formulation

$$\mathrm{Var}(\mathbf{u}) = \frac{1}{N}\sum_{i=1}^{N}\mathbf{u}^\top \mathbf{x}_i \mathbf{x}_i^\top \mathbf{u} = \mathbf{u}^\top \left(\frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i \mathbf{x}_i^\top\right)\mathbf{u}$$

But $\frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i \mathbf{x}_i^\top = \Sigma$

So:

$$\mathrm{Var}(\mathbf{u}) = \mathbf{u}^\top \Sigma \mathbf{u}$$

Step 4: Constrained maximization

We seek $\mathbf{u}_1$ maximizing $\mathbf{u}^\top \Sigma \mathbf{u}$ under $\|\mathbf{u}\| = 1$.

Lagrangian:

$$\mathcal{L}(\mathbf{u}, \lambda) = \mathbf{u}^\top \Sigma \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1)$$

Concise explanation of the role of λ

In PCA, we maximize $\mathbf{u}^T \Sigma \mathbf{u}$ under the constraint $\mathbf{u}^T \mathbf{u} = 1$.
The λ appears naturally when using the **method of Lagrange multipliers** to solve this constrained optimization problem.
The Lagrange function is written:

$$\mathcal{L}(\mathbf{u}, \lambda) = \mathbf{u}^T \Sigma \mathbf{u} - \lambda(\mathbf{u}^T \mathbf{u} - 1)$$

The term $-\lambda(\mathbf{u}^T \mathbf{u} - 1)$ allows us to **incorporate the constraint directly into the objective function**. The multiplier λ serves as a **penalty parameter** that ensures the solution respects the unit norm constraint.
Thus, λ is essentially a **mathematical artifact** that allows us to transform a constrained optimization problem into an equivalent unconstrained problem.

Optimality condition:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{u}} = 2\Sigma \mathbf{u} - 2\lambda \mathbf{u} = 0$$

That is:

$$\Sigma \mathbf{u} = \lambda \mathbf{u}$$

Step 5: Interpretation

For an eigenvector $\mathbf{u}_d$ satisfying $\Sigma \mathbf{u}_d = \lambda_d \mathbf{u}_d$:

$$\mathrm{Var}(\mathbf{u}_d) = \mathbf{u}_d^\top \Sigma \mathbf{u}_d = \mathbf{u}_d^\top (\lambda_d \mathbf{u}_d) = \lambda_d \mathbf{u}_d^\top \mathbf{u}_d = \lambda_d$$

Because $\mathbf{u}_d^\top \mathbf{u}_d = \|\mathbf{u}_d\|^2 = 1$.

Conclusion

Each eigenvalue λ_d is exactly the variance of the data projected onto the corresponding eigenvector $\mathbf{u}_d$.

Order of Components

We order by decreasing eigenvalues:

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_D \geq 0$$

Thus:

- $\mathbf{u}_1$: direction of maximum variance
- $\mathbf{u}_2$: orthogonal direction of remaining maximum variance

Step 4: Component Selection

Explained Variance Criterion

Total Variance

- Represents **all the information** contained in the original data
- This is the **reference**: 100% of available information

$$\text{Total Variance} = \sum_{d=1}^{D} \lambda_d = \mathrm{Tr}(\Sigma)$$

Explained Variance

- Quantifies **the information captured** by each principal component
- Corresponds directly to the **eigenvalues** λ_d of Σ
- The larger λ_d, the more variance the component explains
- The **explained variance** by component d is:

$$\text{ExplainedVar}_d = \frac{\lambda_d}{\sum_{k=1}^{D} \lambda_k}$$

Cumulative Variance

- Progressive sum of explained variances component after component
- Shows **how much information** is retained by selecting K components
- Formula: $\frac{\lambda_1+\lambda_2+\cdots+\lambda_k}{\text{Total variance}}$
- Mathematically, the **cumulative explained variance** for K components is:

$$\text{CumulativeVar}_K = \frac{\sum_{d=1}^{K} \lambda_d}{\sum_{d=1}^{D} \lambda_d}$$

Practical Usefulness

- **Choosing the number of components**: Keep enough components to explain say 80-95% of total variance
- **Dimensionality reduction**: Keep the essential information with fewer variables
- **Interpretation**: Understand which components carry the most information

Visual example: If PC1 explains 60% and PC2 explains 20%, then with 2 components we capture 80% of total information.

Common Selection Rules

- **Elbow Method**: Observe the "elbow" in the eigenvalue plot
- **Variance threshold**: Keep enough components to explain 70-95% of total variance
- **Kaiser criterion**: Keep components with eigenvalue > 1 (if data normalized)

Step 5: Data Projection

The projection of original data onto the first K components is:

$$\mathbf{y}_i = \mathbf{U}_K^\top(\mathbf{x}_i - \overline{\mathbf{x}})$$

where $\mathbf{U}_K = [\mathbf{u}_1, \dots, \mathbf{u}_K]$ contains the first K eigenvectors.

Interpretation of Results

Geometric Interpretation

- **Principal Component 1**: Direction of greatest variance in the data
- **Principal Component 2**: Direction of greatest variance orthogonal to the first
- **Latent space**: New coordinate system optimized for variance

Component Loadings

The elements of the eigenvectors $\mathbf{u}_d$ are the **loadings**:

- **Large absolute value**: Variable strongly contributes to the component
- **Positive/negative sign**: Indicates direction of correlation
- **Interpretation**: Loadings allow understanding which original variables define each component

Observation Scores

The coordinates $\mathbf{y}_i$ in the latent space are the **scores**:

- **Relative position**: Observations with similar scores are similar in original space
- **Visualization**: 2D/3D plot of the first two/three components

Correlation Circle

For normalized data, we can represent:

- **Variables**: Vectors whose length represents quality of representation
- **Angles between vectors**: Correlations between variables
- **Projection on axes**: Correlation with components

Applications and Uses

Dimensionality Reduction

Objective: Go from D dimensions to $K < D$ dimensions with minimal information loss

Reconstruction formula:

$$\tilde{\mathbf{x}}_i = \overline{\mathbf{x}} + \mathbf{U}_K \mathbf{y}_i$$

Reconstruction error:

$$\mathcal{L} = \sum_{i=1}^{N} \|\mathbf{x}_i - \tilde{\mathbf{x}}_i\|^2 = \sum_{d=K+1}^{D} \lambda_d$$

Multidimensional Data Visualization

- **2D/3D projection** for visual exploration
- **Cluster identification** or structure detection
- **Detection of atypical observations** (outliers)

Denoising

If noise has low variance compared to signal, PCA can separate:

- **Signal**: Captured by first components
- **Noise**: Captured by last components

Preprocessing for Other Algorithms

- **Acceleration** of algorithms (fewer dimensions)
- **Performance improvement** (elimination of collinearities)
- **Overfitting reduction**

Limitations and Precautions

PCA Limitations

1. **Linearity**: Does not capture non-linear relationships
2. **Sensitivity to outliers**: Variance is sensitive to extreme values
3. **Difficult interpretation** when components are complex combinations of original variables
4. **Variable scale**: Requires prior normalization
5. **Non-Gaussian distribution**: Less effective for non-normal distributions

Alternatives to Consider

- **Kernel PCA** for non-linear relationships
- **t-SNE, UMAP** for non-linear visualization
- **Autoencoders** for non-linear reduction
- **Factor Analysis** for models with explicit noise

Step-by-Step Application Example

Step 0: Initial Data

Let $\mathbf{X}$ be a data matrix of size $N \times D$

Step 1: Centering and Normalization

```
# Pseudocode for illustration
X_centered = X - mean(X, axis=0)
X_normalized = X_centered / std(X_centered, axis=0)
```

Step 2: Covariance Matrix Calculation

$$\Sigma = \frac{1}{N}\mathbf{X}_{\text{norm}}^{\top}\mathbf{X}_{\text{norm}}$$

Step 3: Eigenvalue Decomposition

Solve $\Sigma \mathbf{u}_d = \lambda_d \mathbf{u}_d$

Step 4: Eigenvalue Analysis

Explained variance table:

Component	Eigenvalue	Explained Variance	Cumulative Variance
PC1	λ_1	$v_1\%$	$v_1\%$
PC2	λ_2	$v_2\%$	$v_1 + v_2\%$
...	...	...	...

Step 5: Decision on Number of Components

- Eigenvalue plot (scree plot)
- Cumulative variance threshold (ex: 80%)
- Loadings analysis for interpretability

Step 6: Projection and Interpretation

- Calculate scores $\mathbf{Y} = \mathbf{X}_{\text{norm}}\mathbf{U}_K$
- Score visualization
- Loadings analysis to name components

Conclusion

Principal Component Analysis is a powerful method but with specific assumptions:

- **Strong power** for linear and Gaussian data
- **Clear geometric interpretation** (variance maximization)
- **Efficient calculation** via eigenvalue decomposition
- **Broad applicability** in preprocessing and visualization

Usage recommendations:

1. Always visualize data before and after PCA
2. Check assumptions (linearity, approximate normality)
3. Normalize variables if they are on different scales
4. Interpret components via loadings
5. Validate results on test data

A thorough understanding of PCA provides a solid foundation for approaching more advanced methods of dimensionality reduction and data analysis.

Choosing the Number of Components in PCA

Question

When performing principal component analysis (PCA), how to choose the number of eigenvectors to keep? Explain why and discuss the implications on reconstruction.

Introduction to the Selection Problem

When we perform Principal Component Analysis, a crucial question arises: **how many principal components should we keep?** This choice represents a fundamental compromise between:

- **Dimensionality reduction** (primary objective)
- **Information preservation** (measured by explained variance)
- **Reconstruction quality** of original data

Main Selection Criteria

Based on Explained Variance

Cumulative Variance Threshold

The most common method is to keep enough components to explain a predefined percentage of total variance.

Calculation:

If we have D eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_D$, total variance is:

$$\text{Total Variance} = \sum_{d=1}^{D} \lambda_d$$

Variance explained by the first K components is:

$$\text{Explained Variance}(K) = \sum_{d=1}^{K} \lambda_d$$

Proportion of explained variance is:

$$\text{PVE}(K) = \frac{\sum_{d=1}^{K} \lambda_d}{\sum_{d=1}^{D} \lambda_d}$$

Practical rule: Choose K such that $\text{PVE}(K) \geq \tau$, where τ is typically between 70% and 95%.

Advantages:

- Simple to understand and implement
- Guarantees minimal information preservation

- Standardized and reproducible

Disadvantages:

- Choice of threshold τ is arbitrary
- Does not account for specific data structure

Eigenvalue Plot Analysis (Scree Plot)

Elbow Method

This visual method consists of plotting eigenvalues as a function of their order.

Procedure:

1. Plot λ_d versus d (for $d = 1, 2, \dots, D$)
2. Identify the point where the curve forms an "elbow" (pronounced angle)
3. Keep components before the elbow

Interpretation:

- Components before the elbow capture the main "signal"
- Components after the elbow mainly capture "noise"

Advantages:

- Intuitive and visual method
- Takes into account relative eigenvalue decay

Disadvantages:

- Subjective (different people may identify different elbows)
- May be ambiguous for some eigenvalue distributions

Kaiser Rule (for Standardized Data)

Eigenvalue $>$ 1 Threshold

When data is standardized (mean 0, variance 1), each original variable contributes 1 to total variance.

Rule: Keep components whose eigenvalue $\lambda_d > 1$.

Justification: A component should explain at least as much variance as a single original variable.

Limitations:

- Only applies to standardized data
- Can be too conservative (too many components) or too lax (too few)

Method Based on Reconstruction Error

Quadratic Error Minimization

A more formal approach is to choose K that minimizes reconstruction error under a constraint.

Reconstruction error: For each observation $\mathbf{x}_i$, reconstruction using K components is:

$$\tilde{\mathbf{x}}_i = \overline{\mathbf{x}} + \sum_{d=1}^{K}(\mathbf{u}_d^\top(\mathbf{x}_i - \overline{\mathbf{x}}))\mathbf{u}_d$$

Total quadratic error is:

$$\mathcal{L}(K) = \sum_{i=1}^{N} \|\mathbf{x}_i - \tilde{\mathbf{x}}_i\|^2 = \sum_{d=K+1}^{D} \lambda_d$$

Important relationship: Reconstruction error is exactly the sum of eigenvalues of excluded components.

Approach: Choose K such that $\mathcal{L}(K) \leq \epsilon$, where ϵ is an acceptable error threshold.

Implications on Reconstruction

Exact Relationship between Number of Components and Quality

Eckart-Young Theorem

This fundamental theorem states that the rank-K approximation by PCA is **optimal** in the least squares sense.

Formalization: Let $\mathbf{X}$ be the centered data matrix. Singular value decomposition (SVD) gives $\mathbf{X} = \mathbf{U}\Sigma\mathbf{V}^\top$. The rank-$K$ approximation:

$$\mathbf{X}_K = \mathbf{U}_K\Sigma_K\mathbf{V}_K^\top$$

minimizes $\|\mathbf{X} - \mathbf{X}_K\|_F^2$ among all rank-K matrices.

Implication: PCA provides the best rank-K approximation for centered data.

Dimension-Reconstruction Trade-off

Extreme Cases

- **$K = D$:** Perfect reconstruction
  - $\tilde{\mathbf{x}}_i = \mathbf{x}_i$ exactly
  - No dimensionality reduction
  - Conservation of 100% variance
- **$K = 0$:** Reconstruction by mean only
  - $\tilde{\mathbf{x}}_i = \overline{\mathbf{x}}$
  - Maximum error: $\mathcal{L}(0) = \sum_{d=1}^{D} \lambda_d$
  - No information about individual variability

Typical Behavior

The relationship between K and reconstruction quality generally follows a **law of diminishing returns**:

- First components significantly improve reconstruction
- Improvements become progressively weaker
- Beyond a certain point, gains are negligible

Impact on Applications

Visualization

For 2D or 3D visualization:

- **$K = 2$ or 3:** Allows direct visualization
- **Implication:** Significant information loss if first two components don't explain enough variance
- **Solution:** Examine several component pairs if necessary

Denoising

PCA can separate signal and noise:

- **Signal:** Captured by first components (high eigenvalues)
- **Noise:** Captured by last components (low eigenvalues)
- **Choice of K:** Should be sufficient to capture signal but exclude noise

Compression

For data compression:

- **Compression rate:** $\frac{K}{D} \times 100\%$
- **Distortion:** Measured by $\frac{\sum_{d=K+1}^{D} \lambda_d}{\sum_{d=1}^{D} \lambda_d} \times 100\%$
- **Trade-off:** Compression rate vs reconstruction quality

Advanced Approaches and Considerations

Cross-Validation

Validation Method

To avoid overfitting, cross-validation can be used:

1. Divide data into training and test sets
2. Calculate PCA on training set
3. Evaluate reconstruction error on test set
4. Choose K minimizing test error

Advantage: More robust, especially for small samples

Information Criteria

Akaike (AIC) and BIC Criteria

For normally distributed data, information criteria can be used:

- **AIC:** Balances fit quality and complexity
- **BIC:** More strongly penalizes complexity

These criteria formulate the choice of K as a model selection problem.

Domain Considerations

Domain Knowledge

Sometimes external considerations guide the choice of K:

- **Interpretability:** Limit to number of components easy to interpret
- **Technical constraints:** Storage or processing limitations
- **Application requirements:** Some applications require specific number of dimensions

Practical Example

Typical Scenario

Suppose analysis on data with $D = 50$ variables and $N = 1000$ observations.

Eigenvalues: $\lambda_1 = 15.2$, $\lambda_2 = 8.1$, $\lambda_3 = 4.3$, $\lambda_4 = 2.1$, $\lambda_5 = 1.8$, … $\lambda_{50} = 0.05$

Calculations:

- Total variance: $\sum \lambda_d = 50.0$ (since standardized data)
- PVE(1): $15.2/50.0 = 30.4\%$
- PVE(2): $(15.2 + 8.1)/50.0 = 46.6\%$
- PVE(3): $(15.2 + 8.1 + 4.3)/50.0 = 55.2\%$
- PVE(4): $(15.2 + 8.1 + 4.3 + 2.1)/50.0 = 59.4\%$
- PVE(5): $(15.2 + 8.1 + 4.3 + 2.1 + 1.8)/50.0 = 63.0\%$

Possible decisions:

1. **80% threshold:** Would require about 10 components (additional calculation needed)
2. **Kaiser rule:** 5 components ($\lambda_5 = 1.8 > 1$, $\lambda_6 = 0.9 < 1$)
3. **Scree plot:** Elbow likely after 3rd or 4th component
4. **Reconstruction error $<$ 40%:** K such that $\sum_{d=K+1}^{50} \lambda_d < 20$ (40% of 50)

General Recommendations

Recommended Approach

1. **Start by standardizing** data if variables have different scales
2. **Examine scree plot** for visual intuition
3. **Calculate cumulative explained variance** for different K
4. **Consider Kaiser rule** if data is standardized
5. **Evaluate your application's needs** in terms of compression and quality
6. **Test robustness** by cross-validation if possible
7. **Check interpretability** of retained components

Pitfalls to Avoid

- **Too few components:** Significant information loss, poor reconstruction
- **Too many components:** Insufficient dimensionality reduction, noise capture
- **Ignoring context:** Optimal choice depends on final application
- **Forgetting validation:** Risk of overfitting on small samples

Conclusion

Choosing the number of components in PCA is a crucial decision that directly influences reconstruction quality and analysis usefulness. No method is universally best; an approach combining several criteria is generally recommended.

Fundamental reminder: Reconstruction error is exactly the sum of eigenvalues of excluded components. This direct link between eigenvalues and approximation quality provides a solid theoretical basis to guide selection.

In practice, this choice always represents a trade-off between simplicity (few dimensions) and fidelity (accurate reconstruction), and must be adapted to the specific objectives of the analysis.